Common Core State Standards Correlations

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Correlation of *Core Connections, Course 2* to the Common Core State Standards Grade 7

This document identifies the lesson(s) in *Core Connections, Course 2* in which primary instruction of each standard from the Common Core State Standards for Mathematics, Grade 7 occurs.

The standards continue to be implemented, applied, and practiced throughout subsequent lessons. #.#.# refers to a lesson in *Core Connections, Course 2*. MN #.#.# refers to the Math Notes box in Lesson #.#.#, Checkpoint # refers to the Checkpoint problems in the back of the student textbook, and #-# refers to an instance of a homework problem in which the standard is implemented. The list of homework problems is by no means comprehensive, but rather intended as a sample.

Standard from CCSS Grade 7	Primary instruction for this standard and a few instances of homework in which the standard is implemented.
Ratios and Proportional Relationships 7.RP	
Analyze proportional relationships and use them to solve real-world and mathematical problems.	
7.RP.1. Compute unit rates associated with ratios of fractions, including ratios of lengths, areas and other quantities measured in like or different units. *For example, if a person walks 1/2 mile in each 1/4 hour, compute the unit rate as the complex fraction 1/2/1/4 miles per hour, equivalently 2 miles per hour.*	4.2.3 MN: 4.2.4 4-61, 4-79, 4-96, 4-118, 4-127, 5-91, 9-92
7.RP.2. Recognize and represent proportional relationships between quantities.	
7.RP.2a. Decide whether two quantities are in a proportional relationship, e.g., by testing for equivalent ratios in a table or graphing on a coordinate plane and observing whether the graph is a straight line through the origin.	4.2.1–4.2.4 4-41, 4-51, 4-77, 4-128, 5-12, 5-103, 6-108, 7-94, 8-78
7.RP.2b. Identify the constant of proportionality (unit rate) in tables, graphs, equations, diagrams, and verbal descriptions of proportional relationships.	4.2.3, 4.2.4 4-60, 5-12, 5-103, 6-108, 6-114, 7-94, 9-50, 9-92
7.RP.2c. Represent proportional relationships by equations. *For example, if total cost t is proportional to the number n of items purchased at a constant price p, the relationship between the total cost and the number of items can be expressed as t = pn.*	4.2.3, 4.2.4, 7.1.1 5-12, 5-103, 8-78, 9-14, 9-50
7.RP.2d. Explain what a point *(x, y)* on the graph of a proportional relationship means in terms of the situation, with special attention to the points (0, 0) and (1, *r)* where *r* is the unit rate.	4.2.2–4.2.4, 7.1.1 4-50, 5-12, 5-103
7.RP.3. Use proportional relationships to solve multistep ratio and percent problems. *Examples: simple interest, tax, markups and markdowns, gratuities and commissions, fees, percent increase and decrease, percent error.*	5.1.1, 5.1.2, 7.1.1, 7.1.3, 7.1.7, 7.1.8, 7.2.1, 7.2.2 5-62, 5-101, 5-139, 5-149, 6-32, 6-82, 8-16, 9-50

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The Number System 7.NS	
Apply and extend previous understandings of operations with fractions to add, subtract, multiply, and divide rational numbers.	
7.NS.1. Apply and extend previous understandings of addition and subtraction to add and subtract rational numbers; represent addition and subtraction on a horizontal or vertical number line diagram.	
7.NS.1a. Describe situations in which opposite quantities combine to make 0. *For example, a hydrogen atom has 0 charge because its two constituents are oppositely charged.*	2.2.2, 2.2.3 MN: 2.2.4 2-63, 2-132, 3-19, 4-117, 6-6, 6-17
7.NS.1b. Understand $p + q$ as the number located a distance $\|q\|$ from p, in the positive or negative direction depending on whether q is positive or negative. Show that a number and its opposite have a sum of 0 (are additive inverses). Interpret sums of rational numbers by describing real-world contexts.	2.2.2, 2.2.3 MN: 2.2.4 2-79, 2-92, 3-35, 3-129
7.NS.1c. Understand subtraction of rational numbers as adding the additive inverse, $p - q = p + (-q)$. Show that the distance between two rational numbers on the number line is the absolute value of their difference, and apply this principle in real-world contexts.	3.2.1, 3.2.2 MN: 3.2.3 3-8, 3-75, 4-6, 4-31, 4-101, 6-8
7.NS.1d. Apply properties of operations as strategies to add and subtract rational numbers.	2.2.2, 2.2.3, 3.1.1, 3.2.1, 3.2.2, 3.3.3 2-53, 2-93, 2-117, 3-20, 3-61, 4-44, 4-53, 4-115, 4-122, 5-87, 7-103
7.NS.2. Apply and extend previous understandings of multiplication and division and of fractions to multiply and divide rational numbers.	
7.NS.2a. Understand that multiplication is extended from fractions to rational numbers by requiring that operations continue to satisfy the properties of operations, particularly the distributive property, leading to products such as $(-1)(-1) = 1$ and the rules for multiplying signed numbers. Interpret products of rational numbers by describing real-world contexts.	2.2.4, 2.2.5, 2.2.6, 3.2.3, 3.2.4 3-110, 3-126, 3-128, 4-101, 4-126, 5-155, 6-59
7.NS.2b. Understand that integers can be divided, provided that the divisor is not zero, and every quotient of integers (with non-zero divisor) is a rational number. If p and q are integers, then $-(p/q) = (-p)/q = p/(-q)$. Interpret quotients of rational numbers by describing real world contexts.	3.2.3, 3.3.1 3-82, 3-83, 3-93, 4-110, 4-126, 6-58, 9-40
7.NS.2c. Apply properties of operations as strategies to multiply and divide rational numbers.	3.2.3–3.3.3 2-101, 2-103, 3-73, 3-94, 3-109, 3-111, 3-136, 3-137, 7-103, 8-37, 8-62
7.NS.2d. Convert a rational number to a decimal using long division; know that the decimal form of a rational number terminates in 0s or eventually repeats.	2.1.1, 2.1.2 2-9, 2-13, 2-26, 2-54, 2-130

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7.NS.3. Solve real-world and mathematical problems involving the four operations with rational numbers. (complex fractions included)	3.2.5, 3.3.1–3.3.3, 7.1.3, 7.1.4 1-39, 2-10, 2-55, 2-66, 3-75, 3-95, 3-127, 7-47, 7-103, 8-62

Expressions and Equations 7.EE	
Use properties of operations to generate equivalent expressions.	
7.EE.1. Apply properties of operations as strategies to add, subtract, factor, and expand linear expressions with rational coefficients.	4.3.1–4.3.3, 6-56 5-53, 5-125, 5-140, 6-59, 6-101, 7-23, 7-62
7.EE.2. Understand that rewriting an expression in different forms in a problem context can shed light on the problem and how the quantities in it are related. *For example, $a + 0.05a = 1.05a$ means that "increase by 5%" is the same as "multiply by 1.05."*	7.1.2–7.1.6 7-33, 7-45, 8-100, 9-35, 9-48
Solve real-life and mathematical problems using numerical and algebraic expressions and equations.	
7.EE.3. Solve multi-step real-life and mathematical problems posed with positive and negative rational numbers in any form (whole numbers, fractions, and decimals), using tools strategically. Apply properties of operations to calculate with numbers in any form; convert between forms as appropriate; and assess the reasonableness of answers using mental computation and estimation strategies. *For example: If a woman making $25 an hour gets a 10% raise, she will make an additional 1/10 of her salary an hour, or $2.50, for a new salary of $27.50. If you want to place a towel bar 9 3/4 inches long in the center of a door that is 27 1/2 inches wide, you will need to place the bar about 9 inches from each edge; this estimate can be used as a check on the exact computation.*	5.3.2–5.3.5, 6.2.3–6.2.5, 7.1.7 6-22, 6-32, 6-86, 6-125, 7-59, 7-61, 7-91
7.EE.4. Use variables to represent quantities in a real-world or mathematical problem, and construct simple equations and inequalities to solve problems by reasoning about the quantities.	
7.EE.4a. Solve word problems leading to equations of the form $px + q = r$ and $p(x + q) = r$, where p, q, and r are specific rational numbers. Solve equations of these forms fluently. Compare an algebraic solution to an arithmetic solution, identifying the sequence of the operations used in each approach. *For example, the perimeter of a rectangle is 54 cm. Its length is 6 cm. What is its width?*	6.2.3–6.2.5, 7.1.6, 7.1.7, Checkpoint 8 (8-109) 6-20, 6-55, 6-81, 6-87, 6-98, 6-124, 6-144, 7-46, 7-72, 8-79, 8-109
7.EE.4b. Solve word problems leading to inequalities of the form $px + q > r$ or $px + q < r$, where p, q, and r are specific rational numbers. Graph the solution set of the inequality and interpret it in the context of the problem. *For example: As a salesperson, you are paid $50 per week plus $3 per sale. This week you want your pay to be at least $100. Write an inequality for the number of sales you need to make, and describe the solutions.*	6.1.4 6-83, 6-92, 6-115, 6-145, 8-39

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Geometry 7.G	
Draw, construct, and describe geometrical figures and describe the relationships between them.	
7.G.1. Solve problems involving scale drawings of geometric figures, including computing actual lengths and areas from a scale drawing and reproducing a scale drawing at a different scale.	4.1.1, 4.1.2, 9.3.2 4-17, 4-84, 5-20, 5-142, 8-46, 9-37, 9-58, 9-67, 9-68
7.G.2. Draw (freehand, with ruler and protractor, and with technology) geometric shapes with given conditions. Focus on constructing triangles from three measures of angles or sides, noticing when the conditions determine a unique triangle, more than one triangle, or no triangle.	8.3.3, 8.3.4 8-85, 8-86, 8-104, 8-105, 9-33, 9-46
7.G.3. Describe the two-dimensional figures that result from slicing three-dimensional figures, as in plane sections of right rectangular prisms and right rectangular pyramids.	9.2.2 9-65, 9-85, 9-89
Solve real-life and mathematical problems involving angle measure, area, surface area, and volume.	
7.G.4. Know the formulas for the area and circumference of a circle and use them to solve problems; give an informal derivation of the relationship between the circumference and area of a circle.	9.1.1–9.1.3 MN: 8.3.3 8-87, 9-10, 9-16, 9-31, 9-32, 9-47, 9-59, 9-78, 9-90, 9-93
7.G.5. Use facts about supplementary, complementary, vertical, and adjacent angles in a multi-step problem to write and solve simple equations for an unknown angle in a figure.	8.3.2 8-74, 8-75, 8-90, 8-111, 8-112, 9-34
7.G.6. Solve real-world and mathematical problems involving area, volume and surface area of two- and three-dimensional objects composed of triangles, quadrilaterals, polygons, cubes, and right prisms.	9.1.3, 9.2.3, 9.2.4 4-17, 4-62, 6-69, 6-100, 9-38, 9-45, 9-60, 9-66, 9-79, 9-94, 9-95

Statistics and Probability 7.SP	
Use random sampling to draw inferences about a population.	
7.SP.1. Understand that statistics can be used to gain information about a population by examining a sample of the population; generalizations about a population from a sample are valid only if the sample is representative of that population. Understand that random sampling tends to produce representative samples and support valid inferences.	8.2.1 MN: 8.2.2 8-35, 9-11, 9-17
7.SP.2. Use data from a random sample to draw inferences about a population with an unknown characteristic of interest. Generate multiple samples (or simulated samples) of the same size to gauge the variation in estimates or predictions. *For example, estimate the mean word length in a book by randomly sampling words from the book; predict the winner of a school election based on randomly sampled survey data. Gauge how far off the estimate or prediction might be.*	8.2.2 8-117, 9-11

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Draw informal comparative inferences about two populations.	
7.SP.3. Informally assess the degree of visual overlap of two numerical data distributions with similar variabilities, measuring the difference between the centers by expressing it as a multiple of a measure of variability. *For example, the mean height of players on the basketball team is 10 cm greater than the mean height of players on the soccer team, about twice the variability (mean absolute deviation) on either team; on a dot plot, the separation between the two distributions of heights is noticeable.*	8.1.1, 8.1.2 8-23, 8-45, 8-63, 9-19
7.SP.4. Use measures of center and measures of variability for numerical data from random samples to draw informal comparative inferences about two populations. *For example, decide whether the words in a chapter of a seventh-grade science book are generally longer than the words in a chapter of a fourth-grade science book.*	8.2.2 8-63, 8-77, 9-82
Investigate chance processes and develop, use, and evaluate probability models.	
7.SP.5. Understand that the probability of a chance event is a number between 0 and 1 that expresses the likelihood of the event occurring. Larger numbers indicate greater likelihood. A probability near 0 indicates an unlikely event, a probability around 1/2 indicates an event that is neither unlikely nor likely, and a probability near 1 indicates a likely event.	1.2.1 2-11, 2-94, 2-15, 5-29, 5-30, 5-61
7.SP.6. Approximate the probability of a chance event by collecting data on the chance process that produces it and observing its long-run relative frequency, and predict the approximate relative frequency given the probability. *For example, when rolling a number cube 600 times, predict that a 3 or 6 would be rolled roughly 200 times, but probably not exactly 200 times.*	1.2.1, 1.2.2, 5.2.3 2-128, 5-61, 5-147, 6-9, 6-33
7.SP.7. Develop a probability model and use it to find probabilities of events. Compare probabilities from a model to observed frequencies; if the agreement is not good, explain possible sources of the discrepancy.	
7.SP.7a. Develop a uniform probability model by assigning equal probability to all outcomes, and use the model to determine probabilities of events. *For example, if a student is selected at random from a class, find the probability that Jane will be selected and the probability that a girl will be selected.*	1.2.3, 5.2.1 4-42, 4-68, 5-38, 5-88, 5-110, 5-126, 5-151
7.SP.7b. Develop a probability model (which may not be uniform) by observing frequencies in data generated from a chance process. *For example, find the approximate probability that a spinning penny will land heads up or that a tossed paper cup will land open-end down. Do the outcomes for the spinning penny appear to be equally likely based on the observed frequencies?*	1.2.2, 5.2.2 5-71, 5-83, 5-86
7.SP.8. Find probabilities of compound events using organized lists, tables, tree diagrams, and simulation.	
7.SP.8a. Understand that, just as with simple events, the probability of a compound event is the fraction of outcomes in the sample space for which the compound event occurs.	1.2.7, 1.2.8, 5.2.5–5.2.6 1-124, 1-125, 2-15, 5-88, 5-110, 7-25, 7-49, 7-125

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7.SP.8b. Represent sample spaces for compound events using methods such as organized lists, tables and tree diagrams. For an event described in everyday language (e.g., "rolling double sixes"), identify the outcomes in the sample space which compose the event.	5.2.4, 5.2.6 5-50, 5-60, 5-72, 5-110, 7-25, 7-101
7.SP.8c. Design and use a simulation to generate frequencies for compound events. *For example, use random digits as a simulation tool to approximate the answer to the question: If 40% of donors have type A blood, what is the probability that it will take at least 4 donors to find one with type A blood?*	5.2.2

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Correlation of the Common Core State Standards Grade 7 to *Core Connections, Course 2*

Chapter 1 Teacher Guide

Section	Lesson	Lesson Objectives	CCSS Standards
1.1	1.1.1	Finding Shared and Unique Characteristics	
	1.1.2	Analyzing a Game	
	1.1.3	Finding Unknowns	
	1.1.4	Investigating a Proportional Relationship	
	1.1.5	Investigating Number Patterns	
1.2	1.2.1	Introduction to Probability	7.SP.5, 7.SP.6
	1.2.2	Investigating Probability	7.SP.6, 7.SP.7b
	1.2.3	Modifying the Sample Space	7.SP.7a
	1.2.4 (opt.)	Expressing Fractions as Percents	
	1.2.5	Rewriting Fractions	Preparation for 7.SP.8a in Lesson 1.2.7
	1.2.6 (opt.)	Fraction Addition	
	1.2.7	Compound Probability	7.SP.8a
	1.2.8	Subtracting Probabilities	7.SP.8a

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Chapter 2 Teacher Guide

Section	Lesson	Lesson Objectives	CCSS Standards
2.1	2.1.1	Fraction-to-Decimal Conversions	7.NS.2d
	2.1.2	Rewriting Decimals as Fractions	7.NS.2d
2.2	2.2.1	Composing Integers	Preparation for 7.NS.1d in Lesson 2.2.2
	2.2.2	Adding Integers and Rational Numbers	7.NS.1a, 7.NS.1b, 7.NS.1d
	2.2.3	More Addition of Integers and Rational Numbers	7.NS.1a, 7.NS.1b, 7.NS.1d
	2.2.4	Multiplication as Repeated Addition	7.NS.2a
	2.2.5	Multiplication of Portions	7.NS.2a
	2.2.6	Multiplying Mixed Numbers	7.NS.2a
2.3	2.3.1	Choosing a Scale and Graphing Data	Preparation for 7.RP.2a, 7.RP.2b in Section 4.2
	2.3.2	More Graph Scaling	Preparation for 7.RP.2a, 7.RP.2b in Section 4.2

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Chapter 3 Teacher Guide

Section	Lesson	Lesson Objectives	CCSS Standards
3.1	3.1.1	Grouping Expressions	7.NS.1d
	3.1.2	Identifying Terms in Expressions	Preparation for 7.NS.1c in Lesson 3.2.1
3.2	3.2.1	Subtraction of Integers	7.NS.1c, 7.NS.1d
	3.2.2	Connecting Addition and Subtraction	7.NS.1c, 7.NS.1d
	3.2.3	Multiplication as Repeated Subtraction	7.NS.2a, 7.NS.2b, 7.NS.2c
	3.2.4	Multiplication with Decimals	7.NS.2a, 7.NS.2c
	3.2.5	Addition, Subtraction, Multiplication, and Division of Integers	7.NS.2c, 7.NS.3
3.3	3.3.1	Division with Rational Numbers	7.NS.2b, 7.NS.2c, 7.NS.3
	3.3.2	Division with Decimals	7.NS.2c, 7.NS.3
	3.3.3	Arithmetic Properties	7.NS.1d, 7.NS.2c, 7.NS.3

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Chapter 4 Teacher Guide

Section	Lesson	Lesson Objectives	CCSS Standards
4.1	4.1.1	Similar Figures	7.G.1
	4.1.2	Scale Drawings	7.G.1
4.2	4.2.1	Recognizing Proportional Relationships	7.RP.2a
	4.2.2	Proportional Relationships with Tables and Graphs	7.RP.2a, 7.RP.2d
	4.2.3	Unit Rate and Proportional Equations	7.RP.1, 7.RP.2a, 7.RP.2b, 7.RP.2c, 7.RP.2d
	4.2.4	Connecting Representations of Proportional Relationships	7.RP.2a, 7.RP.2b, 7.RP.2c, 7.RP.2d
4.3	4.3.1	Combining Like Terms	7.EE.1
	4.3.2	Distributive Property	7.EE.1
	4.3.3	Simplifying with Zero	7.EE.1

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Chapter 5 Teacher Guide

Section	Lesson	Lesson Objectives	CCSS Standards
5.1	5.1.1	Part-Whole Relationships	7.RP.3
	5.1.2	Finding and Using Percentages	7.RP.3
5.2	5.2.1	Probability Games	7.SP.7a
	5.2.2	Computer Simulations of Probability	7.SP.7b, 7.SP.8c
	5.2.3	Compound Independent Events	7.SP.6
	5.2.4	Probability Tables	7.SP.8b
	5.2.5	Probability Trees	7.SP.8a
	5.2.6	Compound Events	7.SP.8a, 7.SP.8b
5.3	5.3.1	Describing Relationships Between Quantities	Preparation for 7.EE.3 in Lesson 5.3.2
	5.3.2	Solving a Word Problem	7.EE.3
	5.3.3	Strategies for Using the 5-D Process	7.EE.3
	5.3.4	Using Variables to Represent Quantities in Word Problems	7.EE.3
	5.3.5	More Word Problem Solving	7.EE.3

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Chapter 6 Teacher Guide

Section	Lesson	Lesson Objectives	CCSS Standards
6.1	6.1.1	Comparing Expressions	Preparation for 7.EE.4a, 7.EE.4b in Lessons 6.1.4, 6.2.1
	6.1.2	Comparing Quantities with Variables	Preparation for 7.EE.4a, 7.EE.4b in Lessons 6.1.4, 6.2.1
	6.1.3	One Variable Inequalities	Preparation for 7.EE.4a, 7.EE.4b in Lessons 6.1.4, 6.2.1
	6.1.4	Solving One Variable Inequalities	7.EE.4b
6.2	6.2.1	Solving Equations	7.EE.4a
	6.2.2	Checking Solutions and the Distributive Property	7.EE.4a
	6.2.3	Solving Equations and Recording Work	7.EE.3, 7.EE.4a
	6.2.4	Using a Table to Write Equations from Word Problems	7.EE.3, 7.EE.4a
	6.2.5	Writing and Solving Equations	7.EE.3, 7.EE.4a
	6.2.6	Cases With Infinite or No Solutions	7.EE.4a
	6.2.7	Choosing a Solving Strategy	7.EE.4a

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Chapter 7 Teacher Guide

Section	Lesson	Lesson Objectives	CCSS Standards
7.1	7.1.1	Distance, Rate, and Time	7.RP.2c, 7.RP.2d, 7.RP.3
	7.1.2	Scaling Quantities	7.EE.2
	7.1.3	Solving Problems Involving Percents	7.NS.3, 7.EE.2, 7.RP.3
	7.1.4	Equations with Fraction and Decimal Coefficients	7.NS.3, 7.EE.2
	7.1.5	Creating Integer Coefficients	7.EE.2
	7.1.6	Creating Integer Coefficients Efficiently	7.EE.2, 7.EE.4a
	7.1.7	Percent Increase and Decrease	7.RP.3, 7.EE.3, 7.EE.4a
	7.1.8	Simple Interest	7.RP.3
7.2	7.2.1	Finding Missing Information in Proportional Relationships	7.RP.3
	7.2.2	Solving Proportions	7.RP.3

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Chapter 8 Teacher Guide

Section	Lesson	Lesson Objectives	CCSS Standards
8.1	8.1.1	Measurement Precision	7.SP.3
	8.1.2	Comparing Distributions	7.SP.3
8.2	8.2.1	Representative Samples	7.SP.1
	8.2.2	Inference From Random Samples	7.SP.2, 7.SP.4
8.3	8.3.1	Introduction to Angles	Preparation for 7.G.5 in Lesson 8.3.2
	8.3.2	Classifying Angles	7.G.5
	8.3.3	Constructing Shapes	7.G.2
	8.3.4	Building Triangles	7.G.2

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Chapter 9 Teacher Guide

Section	Lesson	Lesson Objectives	CCSS Standards
9.1	9.1.1	Circumference, Diameter, and Pi	7.G.4
	9.1.2	Area of Circles	7.G.4
	9.1.3	Area of Composite Shapes	7.G.4, 7.G.6
9.2	9.2.1	Surface Area and Volume	7.G.6
	9.2.2	Cross Sections	7.G.3
	9.2.3	Volume of a Prism	7.G.6
	9.2.4	Volume of Non-Rectangular Prisms	7.G.6

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CCSS Mathematical Practices in CPM Courses

All lessons in a CPM course integrate the CCSS Mathematical Practices. CPM does not treat the CCSS Mathematical Practices as an occasional activity, nor does it simply tack them onto the lessons. The CCSS Mathematical Practices follow, along with descriptions of how they are implemented throughout the CPM curriculum.

1. Make sense of problems and persevere in solving them.
Mathematically proficient students start by explaining to themselves the meaning of a problem and looking for entry points to its solution. They analyze givens, constraints, relationships, and goals. They make conjectures about the form and meaning of the solution and plan a solution pathway rather than simply jumping into a solution attempt. They consider analogous problems, and try special cases and simpler forms of the original problem in order to gain insight into its solution. They monitor and evaluate their progress and change course if necessary. Older students might, depending on the context of the problem, transform algebraic expressions or change the viewing window on their graphing calculator to get the information they need. Mathematically proficient students can explain correspondences between equations, verbal descriptions, tables, and graphs or draw diagrams of important features and relationships, graph data, and search for regularity or trends. Younger students might rely on using concrete objects or pictures to help conceptualize and solve a problem. Mathematically proficient students check their answers to problems using a different method, and they continually ask themselves, "Does this make sense?" They can understand the approaches of others to solving complex problems and identify correspondences between different approaches.

The CPM courses have students solve realistic, non-routine problems that are rich in mathematics on a daily basis. The guided investigations are not merely a set of "word problems" that mimic algorithmic, computational problems. By having students make sense of each problem, rather than being told how to solve a particular kind of problem step by step, the CPM curriculum helps students to develop deep conceptual understanding of the mathematics, procedural fluency, and perseverance. In addition, the curriculum teaches problem-solving strategies that equip students with strategic competence and adaptive reasoning.

2. Reason abstractly and quantitatively.
Mathematically proficient students make sense of quantities and their relationships in problem situations. They bring two complementary abilities to bear on problems involving quantitative relationships: the ability to *decontextualize*—to abstract a given situation and represent it symbolically and manipulate the representing symbols as if they have a life of their own, without necessarily attending to their referents—and the ability to *contextualize*, to pause as needed during the manipulation process in order to probe into the referents for the symbols involved. Quantitative reasoning entails habits of creating a coherent representation of the problem at hand; considering the units involved; attending to the meaning of quantities, not just how to compute them; and knowing and flexibly using different properties of operations and objects.

Instead of offering word problems only at the end of each chapter or problem set, the CPM curriculum generally presents mathematical ideas in contexts *first*, helping students make sense of otherwise abstract principles. Only then do students move on to abstraction and generalization using symbolic notation. Students are taught to try specific numbers to evaluate what is happening in problems, critically analyze units, identify what numbers or solutions mean in a given context, write number sentences or expressions that represent given relationships, and identify patterns and relationships that lead to solutions.

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3. Construct viable arguments and critique the reasoning of others.
Mathematically proficient students understand and use stated assumptions, definitions, and previously established results in constructing arguments. They make conjectures and build a logical progression of statements to explore the truth of their conjectures. They are able to analyze situations by breaking them into cases, and can recognize and use counterexamples. They justify their conclusions, communicate them to others, and respond to the arguments of others. They reason inductively about data, making plausible arguments that take into account the context from which the data arose. Mathematically proficient students are also able to compare the effectiveness of two plausible arguments, distinguish correct logic or reasoning from that which is flawed, and—if there is a flaw in an argument—explain what it is. Elementary students can construct arguments using concrete referents such as objects, drawings, diagrams, and actions. Such arguments can make sense and be correct, even though they are not generalized or made formal until later grades. Later, students learn to determine domains to which an argument applies. Students at all grades can listen or read the arguments of others, decide whether they make sense, and ask useful questions to clarify or improve the arguments.

In CPM courses, students regularly share information, opinions, and their knowledge and understandings in study teams. They contribute, listen, argue, ask each other for help, check for understanding, and keep each other focused. During this process students use higher-order thinking skills by providing clarification, building on the ideas of others, analyzing and coming to consensus. Justifying and critiquing happens every day in a CPM classroom; it does not occur as an occasional assignment. Students must communicate their mathematical findings in a clear and convincing manner, via narratives, oral presentations, or posters. Teachers answer students' questions, doing so in a way that challenges and motivates students to conjecture and test solutions themselves.

4. Model with mathematics.
Mathematically proficient students can apply the mathematics they know to solve problems arising in everyday life, society, and the workplace. In early grades, this might be as simple as writing an addition equation to describe a situation. In middle grades, a student might apply proportional reasoning to plan a school event or analyze a problem in the community. By high school, a student might use geometry to solve a design problem or use a function to describe how one quantity of interest depends on another. Mathematically proficient students who can apply what they know are comfortable making assumptions and approximations to simplify a complicated situation, realizing that these may need revision later. They are able to identify important quantities in a practical situation and map their relationships using such tools as diagrams, two-way tables, graphs, flowcharts and formulas. They can analyze those relationships mathematically to draw conclusions. They routinely interpret their mathematical results in the context of the situation and reflect on whether the results make sense, possibly improving the model if it has not served its purpose.

Modeling contextual situations with multiple representations recurs throughout the CPM curriculum. From their first encounters with basic functions like proportions and linear equations, students learn to model situations using tables, graphs, equations, written descriptions, and diagrams. This approach continues throughout each course, but with increasingly complex functions. In doing so, students make assumptions, form predictions, and then check to see if their predictions make sense in the context of the problem. Students use area models to multiply fractions, multiply and divide polynomials, factor, and solve probability problems. Students learn that models may not be perfect, yet can be very useful for describing data and making predictions.

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5. Use appropriate tools strategically.
Mathematically proficient students consider the available tools when solving a mathematical problem. These tools might include pencil and paper, concrete models, a ruler, a protractor, a calculator, a spreadsheet, a computer algebra system, a statistical package, or dynamic geometry software. Proficient students are sufficiently familiar with tools appropriate for their grade or course to make sound decisions about when each of these tools might be helpful, recognizing both the insight to be gained and their limitations. For example, mathematically proficient high school students analyze graphs of functions and solutions generated using a graphing calculator. They detect possible errors by strategically using estimation and other mathematical knowledge. When making mathematical models, they know that technology can enable them to visualize the results of varying assumptions, explore consequences, and compare predictions with data. Mathematically proficient students at various grade levels are able to identify relevant external mathematical resources, such as digital content located on a website, and use them to pose or solve problems. They are able to use technological tools to explore and deepen their understanding of concepts.

In a typical CPM lesson, students have the option of using various tools—including constructing physical models, drawing diagrams, using manipulatives, using calculators, or using eTools—but are not typically told which specific tools to use to solve any particular problem. It is not unusual for different teams to use different tools to solve a problem. During a typical lesson, students share their solution strategies with the whole class, which frequently provokes a lively discussion of which tools were most efficient and productive to solve a given problem.

6. Attend to precision.
Mathematically proficient students try to communicate precisely to others. They try to use clear definitions in discussion with others and in their own reasoning. They state the meaning of the symbols they choose, including using the equal sign consistently and appropriately. They are careful about specifying units of measure, and labeling axes to clarify the correspondence with quantities in a problem. They calculate accurately and efficiently, express numerical answers with a degree of precision appropriate for the problem context. In the elementary grades, students give carefully formulated explanations to each other. By the time they reach high school they have learned to examine claims and make explicit use of definitions.

When using the CPM curriculum, students must communicate clearly, giving carefully formulated explanations to each other, hence they understand the need for attending to precision. While working on problems, students pay attention to units and check whether a numerical solution makes sense. Other times they are precise in scaling or reading the scale on a graph. The use of calculators in some CPM investigations requires students to attend to the precision of the results displayed.

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7. Look for and make use of structure.
Mathematically proficient students look closely to discern a pattern or structure. Young students, for example, might notice that three and seven more is the same amount as seven and three more, or they may sort a collection of shapes according to how many sides the shapes have. Later, students will see 7×8 equals the well remembered $7 \times 5 + 7 \times 3$, in preparation for learning about the distributive property. In the expression $x^2 + 9x + 14$, older students can see the 14 as 2×7 and the 9 as $2 + 7$. They recognize the significance of an existing line in a geometric figure and can use the strategy of drawing an auxiliary line for solving problems. They also can step back for an overview and shift perspective. They can see complicated things, such as some algebraic expressions, as single objects or as being composed of several objects. For example, they can see $5 - 3(x - y)^2$ as 5 minus a positive number times a square and use that to realize that its value cannot be more than 5 for any real numbers x and y.

Since CPM students are developing conceptual understanding of the underlying mathematics, they frequently look for and make use of structure. For example, area models are used to understand multiplication and division, or factoring problems. Giant Ones—fractions where the numerator and denominator are equal—are used to simplify rational expressions, using the underlying structure of the expression instead of just 'canceling out'. In the middle school courses, the 5-D Process is used to understand the structure of the expression needed to solve a problem. In the topic of geometry, students use the structures of polygons to classify them and later prove theorems about them.

8. Look for and express regularity in repeated reasoning.
Mathematically proficient students notice if calculations are repeated, and look both for general methods and for shortcuts. Upper elementary students might notice when dividing 25 by 11 that they are repeating the same calculations over and over again, and conclude they have a repeating decimal. By paying attention to the calculation of slope as they repeatedly check whether points are on the line through (1, 2) with slope 3, middle school students might abstract the equation $(y - 2)/(x - 1) = 3$. Noticing the regularity in the way terms cancel when expanding $(x - 1)(x + 1)$, $(x - 1)(x^2 + x + 1)$, and $(x - 1)(x^3 + x^2 + x + 1)$ might lead them to the general formula for the sum of a geometric series. As they work to solve a problem, mathematically proficient students maintain oversight of the process, while attending to the details. They continually evaluate the reasonableness of their intermediate results.

In a CPM course, students often look for and generalize patterns. Patterns are used repeatedly to recognize generalizations or make conjectures about rules. Repeated reasoning allows for increasingly complex problems to be solved from simpler ones. For example, students use patterns to make sense of negative, zero, and fractional exponents. They also recognize patterns when factoring different expressions.

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Integration of the CCSS Mathematical Practices in a Sample Lesson

A typical CPM lesson reveals depth of integration of the CCSS Mathematical Practices into the course. In Lesson 9.1.1 of CPM's *Core Connections, Course 2*, students work in collaborative teams and as a class to discover the proportional relationship between the diameter and circumference of a circle. The lesson launches (problem 9-1, "Bubble Madness,") with the teacher asking students to predict how the length of a string wrapped around a can of tennis balls would compare to the height of the can. Students make predictions by imagining and visualizing (Practice 2), not by actually measuring, and then share predictions in a brief class discussion (Practice 3). Students then move into the main activity for the day, which will help them answer this question (Practice 1). In problem 9-2, the students work in teams to gather data (Practice 4) in order to determine the relationship between the circumference of a circle and its diameter. They do this by blowing bubbles with a soap solution and then letting the bubbles pop on a piece of construction paper, which leaves a clear circle. Students then measure the distance around and across the circle with string and a ruler or meter stick (Practices 5 and 6). Once they have gathered data for at least 8 different circles students organize their data in a table and create an appropriate graph (Practices 4 and 5). In order to make the graph students need to locate tenths of centimeters, which requires them to reason quantitatively (Practice 2). In problem 9-3, students turn to interpreting the pattern or relationship in the data on their graph (Practices 7 and 8). It is particularly important in this step for students to recall that their graph is based on measurements and will necessarily include a degree of error (Practice 6). The teacher leads a whole class discussion about whether the two measurements are related proportionally, drawing upon knowledge the students have developed in previous lessons regarding the characteristics of proportional relationships in tables and graphs (Practices 2 and 3). In the remainder of the lesson students connect their investigation discoveries to formal definitions and measurements of π (Practice 6) and work to generalize the relationship between circumference and diameter (Practice 2). In problem 9-7, students return to the opening question, decide if their prediction has changed, and discuss this with their team (Practice 3). Then they test their prediction (Practices 4 and 5). The final problem of the lesson, problem 9-8, contains applications and extensions of what students have learned about circumference and diameter (Practice 7). For the "Closure" activity of the lesson, the class discusses and then each student formally records the generalization of the relationship (Practices 3, 4, and 7).

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Correlation of *Core Connections, Course 2* to the CCSS Mathematical Practices

The table below lists sample lessons that the reader can review to see how CCSS Mathematical Practices are embedded in all the lessons. The reader should examine the detailed sample lesson on the previous page before going through any of the lessons on the next page. The table is by no means exhaustive; it illustrates only how the practices are integrated into a few typical lessons. The CCSS Mathematical Practices stand as the foundation of *each* of the daily lessons. An "xx" in the table represents a practice that is a focus of the lesson. An "x" represents a practice that is present in the lesson.

In addition to the daily execution of the CCSS Mathematical Practices as shown in the table below, culminating problems at the end of the course ask students to specifically reflect on when, where, and how they used the CCSS Mathematical Practices during the course (problems 9-104, 9-107, 9-111).

	Mathematical Practice							
Lesson #.#.# Title	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
2.2.1 Composing Integers		x					xx	
2.3.1 Choosing a Scale and Graphing Data						xx		
3.1.1 Grouping Expressions							xx	
3.2.2 Connecting Addition and Subtraction		xx				x		
4.1.2 Scale Drawings				xx				
4.2.2 Proportional Relationships with Tables and Graphs	x	x		x	x		x	
4.3.1 Combining Like Terms					xx		x	
5.2.4 Probability Tables	xx	x	x		x			
5.3.2 Solving a Word Problem	x			x	xx	x		
6.2.1 Solving Equations		xx	x				x	
6.2.4 Use a Table to Write Equations from Word Problems	x			xx		x		
7.1.2 Scaling Quantities	xx			x		x		
7.2.1 Finding Missing Information in Proportional Relationships		x				xx		
8.1.1 Measurement Precision	x	x			x	xx		
8.3.1 Introduction to Angles		xx		x		x		
9.1.1 Circumference, Diameter, Pi		xx			x			xx
9.2.2 Cross Sections		x	xx					

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